FAQ-10: How Do I Size and Scale ActiveGates?

> **Series:** FAQ — Frequently Asked Questions | **Reference:** 10 — ActiveGate Sizing and Scaling | **Created:** July 2026 | **Last Updated:** 10/22/2026

Overview

"How big should my ActiveGate be?" is really three questions wearing one coat: **what jobs** will this ActiveGate do, **how much volume** does each job carry, and **how much headroom** do you need for failover and growth. Dynatrace publishes baseline sizing for each job — but the numbers live on different documentation pages (host-based routing, containerized Kubernetes monitoring, private synthetic locations), and none of them tells you when an ActiveGate you already run has outgrown its size.

This entry consolidates the decision: which dimension drives sizing for each ActiveGate capability, the current documented baselines for host-based, containerized, and synthetic-enabled ActiveGates, how high availability and network zones change the math, and — the part teams most often skip — the `dt.sfm.active_gate.*` self-monitoring metrics that tell you an ActiveGate is saturating *before* data starts dropping.

Deployment mechanics stay with their owning series: ONBRD-03 (installing and deploying ActiveGates), the K8S series (DynaKube and containerized ActiveGates), SYNTH (private synthetic locations), and FAQ-05 (managing ActiveGate updates). This entry owns the sizing decision itself.

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Prerequisites

Requirement	Details
Audience	Platform engineers, Dynatrace administrators, and architects planning or operating ActiveGate fleets
Format	Decision-support document — consolidates sizing baselines and scaling signals; not a hands-on installation lab
Deployment	Dynatrace SaaS; Environment ActiveGates (host-based and/or containerized)
Related topic series	ONBRD-03 (deploying ActiveGate — installation, groups, example deployment), K8S (DynaKube and containerized ActiveGate mechanics), SYNTH-04 (private synthetic locations), M2S-03 / S2S-04 (ActiveGate design during migration)
Related FAQ	FAQ-05 (managing ActiveGate updates on SaaS — roles, sequencing, update rings)

1. Short Answer

Size an ActiveGate in three steps:

1. **Decide what the ActiveGate will do.** Each capability has its own sizing dimension — you cannot size an ActiveGate without knowing its job list, and demanding capabilities (synthetic, Kubernetes platform monitoring) deserve dedicated ActiveGates rather than sharing one box.
2. **Start from the documented baseline for that dimension.** Hosts routed → the host-based tables (§3). Pods monitored → the Kubernetes S/M/L scenarios (§4). Browser monitor executions per hour → the synthetic node sizes (§5).
3. **Provision headroom and verify with self-monitoring.** Dynatrace's guidance is that the ActiveGate machine *"should not exceed 50% CPU and 80% memory"* — treat those as your alert thresholds, watch the self-monitoring metrics (§7), and scale out for routing or up for Kubernetes monitoring when you approach them (§8).

The capability-to-dimension map:

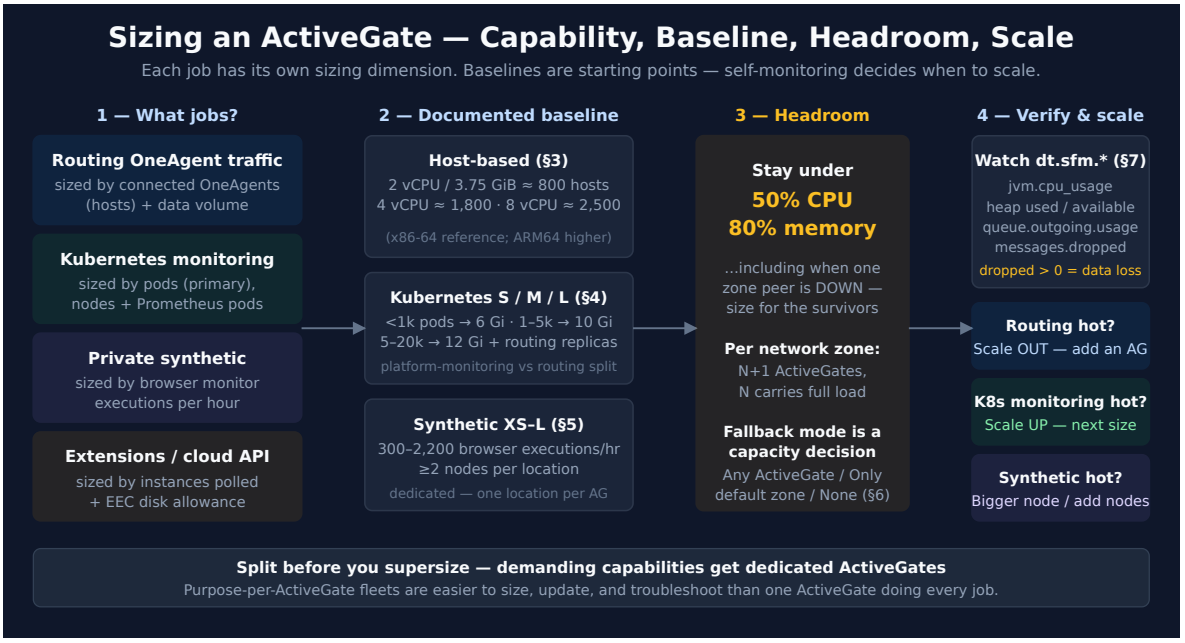
ActiveGate job	What drives its sizing	Baseline
Routing OneAgent traffic	Number of connected OneAgents (hosts) and their data volume	§3
Kubernetes monitoring	Pod count (primary), node count (secondary), Prometheus-annotated pods	§4

ActiveGate job	What drives its sizing	Baseline
Cloud & remote API monitoring (AWS, Azure, VMware, SNMP, Prometheus…)	Number of monitored endpoints/entities polled per cycle	§3 + extension notes
Extensions (EEC)	Extension instances and monitored devices; adds disk requirements	§3
Private synthetic	Browser monitor executions per hour (HTTP monitors are far cheaper)	§5
z/OS routing	Mainframe traffic via the zRemote module	Size with Dynatrace guidance for your mainframe volume

One rule sits above all the tables: **sizing numbers are starting points, not guarantees.** Actual capacity depends on your data volume per host, extension mix, and network — which is why §7 (measuring the real utilization) is part of sizing, not an afterthought.

> **Scope — Dynatrace SaaS.** This entry covers **Environment ActiveGates on SaaS**. On Dynatrace Managed the Environment-ActiveGate baselines broadly transfer (same machines, same jobs), but Managed adds two sizing domains this entry deliberately doesn't cover — **Cluster ActiveGates** (the proxies, usually in a DMZ, that let OneAgents, RUM beacons and synthetic traffic from outside your network reach a Managed cluster) and the sizing of the Managed cluster nodes themselves — and §7's Grail-native `dt.sfm.*` metrics aren't available there, leaving the classic `dsfm:` family as the self-monitoring surface. For Cluster ActiveGate placement and health on Managed, see DTM-14 — §9.4 covers sizing, which the docs do not document as a table. DTM-04 §4 covers ActiveGate redundancy and sizing on Managed, and DTM-15 the sizing of the cluster nodes. If you're consolidating from Managed to SaaS, the M2S series owns that journey.
>
> **Sources:** Linux ActiveGate hardware and system requirements (DT docs), Dynatrace ActiveGate (DT docs) — incl. the Environment vs Cluster ActiveGate split, Managed deployments (DT docs) — *"use one or more Cluster ActiveGates as mediating proxies for pre-processing of OneAgent and DEM traffic."* **Derived:** the three-step framing and the capability-to-dimension map synthesize the per-capability sizing pages cited throughout this entry; the Managed scope note combines the hub's ActiveGate-type split with §7's Grail-bucket residency (Managed environments don't include Grail).

2. What Drives ActiveGate Load



The ActiveGate hub enumerates its jobs: routing OneAgent traffic as a secure proxy, monitoring cloud environments and remote technologies via API (AWS, VMware, Azure, Kubernetes, OpenShift, Google Cloud, SNMP, Prometheus, and more), running synthetic monitors from private locations, routing z/OS traffic, and providing Dynatrace API access. Each job consumes a different resource profile:

- **Routing** is network- and CPU-bound: load scales with the number of OneAgents connected and the volume of data they send. Memory matters for the outgoing message queue when the uplink is slow.
- **Kubernetes monitoring** is memory- and CPU-bound on the *monitored workload count*: Dynatrace's Kubernetes sizing guide is explicit that consumption *"scales with the number of pods primarily due to increased data processing and storage needs,"* with node count a secondary driver and Prometheus metric collection adding CPU as annotated-pod counts rise.
- **Extensions (EEC)** add per-instance polling work plus disk: the Extension Execution Controller wants ~2.1 GB of free disk for its persistence mechanism, and the extensions module claims its own installation and log/cache space.
- **Synthetic** is the most demanding per unit of work — real browsers rendering real pages. Dynatrace states plainly that *"a Synthetic-enabled ActiveGate has more demanding hardware and system requirements than a regular Environment or Cluster ActiveGate."*

Two placement consequences follow. First, Dynatrace recommends a **dedicated system** — *"For optimal performance and enhanced security, we recommend installing and running ActiveGate on a dedicated system."* Second, **don't stack demanding capabilities on one ActiveGate** — the Kubernetes guide's recommended production pattern is itself a split (one ActiveGate for platform monitoring, separate replicas for OneAgent routing, §4), and synthetic locations can't share an ActiveGate with other locations at all (§5). A routing-only ActiveGate plus purpose-built ActiveGates for heavy jobs is easier to size, easier to diagnose, and safer to update (FAQ-05's update-ring guidance assumes the same separation).

> **Sources:**
>
> - Dynatrace ActiveGate (DT docs)
> - Sizing guide for ActiveGates in the Kubernetes monitoring use-case (DT docs)
> - Requirements for private Synthetic locations (DT docs)
> - Linux ActiveGate hardware and system requirements (DT docs) — *"For optimal performance and enhanced security, we recommend installing and running ActiveGate on a dedicated system."*
> - **Derived:** the per-capability resource-profile characterizations combine the cited sizing drivers with the deployment-separation recommendations

3. Baseline Sizing — Host-Based ActiveGates

Minimums (routing/monitoring ActiveGate): 2 GB RAM (4 GB recommended), one dual-core processor, 4 GB free disk for installation/configuration/logs plus 4 GB for cached OneAgent installers and container images if the ActiveGate serves them.

Capacity by machine size. Dynatrace publishes estimated host counts per reference instance size, and the machine *"should not exceed 50% CPU and 80% memory"* in steady state:

Architecture	Reference instance	vCPU	RAM (GiB)	Estimated hosts routed
x86-64	c6i.large	2	3.75	~800
x86-64	c6i.xlarge	4	7.5	~1,800
x86-64	c6i.2xlarge	8	15	~2,500
ARM64	c7g.large	2	3.75	~1,300
ARM64	c7g.xlarge	4	7.5	~2,700
ARM64	c7g.2xlarge	8	15	~5,500
s390	S	2	4	~800
s390	M	4	8	~1,500

Reading the table honestly: the reference points are AWS instance shapes, so for on-premises VMs map by vCPU/RAM, and treat "estimated hosts" as exactly that — an estimate at typical per-host data volume. Hosts running chatty workloads (heavy log routing through the ActiveGate, dense process instrumentation) consume capacity faster.

Disk beyond the minimums. The requirements page itemizes where space goes — installation (~600 MB executables), logs (1.2 GB), auto-update downloads (600 MB), temporary files (4 GB) — and Environment ActiveGates running extensions add 2 GB for the extensions module plus 2 GB for its logs/cache. The EEC's log persistence wants a further 2136 MB of free disk space — 600 MB for the reliability mechanism plus 1.5 GB as a buffer. Without it, *"the persistence will be turned off and log*

ingestion will be transmitted without the reliability mechanism": extensions keep sending, but lose the buffer that covers an outage. A practical floor for an extensions-running ActiveGate is ~15 GB of free disk; disk is cheap, resizing later is not.

Windows ActiveGates follow the same sizing model with their own OS-support matrix — see the Windows requirements page alongside the Linux one.

> **Update (SaaS 1.343, July 2026):** Dynatrace published official **sizing guides for Environment ActiveGates** with capacity recommendations specifically for **log ingestion** infrastructure, plus an upgraded **log ingest monitoring dashboard** covering OneAgent modules and Environment ActiveGate. If your AG fleet fronts log ingestion, size against those guides first — they postdate (and refine) the general host-based baselines in this section for that workload.
>
> The host-based guide is the *Sizing guide for API log ingestion only* section of the Linux ActiveGate requirements page (DT docs) — an ActiveGate receiving logs exclusively through the Log ingestion API, at a typical message-size mix:
>
> | Instance | vCPU | RAM (GiB) | API log ingestion only (MB/min) |
> |---|---|---|---|
> | c7i.large | 2 | 3.75 | 1,100 |
> | c7i.xlarge | 4 | 7.5 | 2,300 |
> | c7i.2xlarge | 8 | 15 | 5,100 |
> | c7g.large | 2 | 3.75 | 1,000 |
> | c7g.xlarge | 4 | 7.5 | 2,000 |
> | c7g.2xlarge | 8 | 15 | 4,600 |
>
> The same page carries a separate guide for a combined workload (OneAgent routing traffic plus API log ingestion); use that one if the ActiveGate does both.
>
> **Sources:** Linux ActiveGate hardware and system requirements (DT docs) — sizing tables, 50%/80% guideline, disk itemization, and EEC persistence figures quoted from this page; the log-ingestion table is its *Sizing guide for API log ingestion only* section (*"ActiveGate receives log data exclusively through the Log ingestion API."*). SaaS 1.343 release notes (DT docs) — *"Sizing guides for Environment ActiveGate Log ingestion API"*. **Derived:** the ~15 GB practical disk floor sums the itemized directories with growth margin; the "chatty workloads consume capacity faster" caveat generalizes the per-host-volume dependency the docs imply but don't quantify.

4. Sizing Containerized ActiveGates on Kubernetes

Containerized ActiveGates (deployed via DynaKube — mechanics in the K8S series) size against the **monitored cluster's workload count**, not host counts. Dynatrace's Kubernetes sizing guide sets three scenarios:

Scenario	Pods monitored	CPU request	CPU limit	Memory request = limit
S	< 1,000	200m	1000m	6 Gi
M	1,000–5,000	1000m	2000m	10 Gi
L	5,000–20,000	2000m	4000m	12 Gi

Operational notes straight from the guide:

- **Prefer no CPU limits** where cluster policy allows — throttling an ActiveGate delays its processing pipeline.
- **Split platform monitoring from routing.** The guide recommends *"to run two sets of ActiveGates for production deployments"*: one for Kubernetes platform monitoring, one for OneAgent traffic routing and telemetry ingest. The platform-monitoring table above gives one requests/limits set per cluster size and no replica count; the DynaKube reference documents `kubernetesMonitoring.replicas` for high availability (*"Set more than one replica for high-availability mode."*), so capacity for that workload comes from moving up a size. The routing set carries a replica count per size:

Pods	Routing CPU request/limit	Memory request = limit	Replicas
< 1,000	250m / 1000m	2 Gi	3
1,000–5,000	500m / 2000m	4 Gi	3
5,000–20,000	1000m / 4000m	6 Gi	6

Log ingest on Kubernetes. SaaS 1.343 extended the ActiveGate log-ingest sizing guides to containerized deployments (*"These now cover containerized (operator-provisioned) and host-based deployments."*). The guide now sizes ActiveGates that receive OTLP logs, per replica:

Cluster size	CPU request / limit	Memory request = limit	OTLP log throughput per replica
Small	250m / 1000m	2 GiB	350 MB/min
Medium	500m / 2000m	4 GiB	650 MB/min
Large	1000m / 4000m	6 GiB	1,200 MB/min

For an ActiveGate that does both OneAgent routing and OTLP log ingest, start from the routing table above and add *"additional replicas = ceil(expected log traffic (MB/min) ÷ throughput per replica)"*. For **Log Agent** traffic routing, the guide's benchmark is 750 MB/min sustained per replica at 500m / 2000m CPU and 4Gi memory. And it names the point to stop growing one deployment: *"If you observe consistent resource exhaustion (CPU usage >80%, memory usage >85%, or frequent GC pauses), consider splitting the workload into separate ActiveGate deployments rather than continuously increasing resources for a single deployment."*

The guide also publishes its own health thresholds — effectively the containerized version of §7: raise the CPU request when usage exceeds **85%** of it or CPU throttling exceeds **10%**; raise the memory request when the working set passes **80%**; and treat a Kubernetes-monitoring pipeline execution time beyond **50–60 seconds** as a CPU-shortage signal.

> **ActiveGate 1.343 and 1.345 — a pod-scope reading can move without the workload changing.** Both releases change how the Kubernetes CPU and memory **request and limit** metrics are reported, and both are narrower than they first look:
>
> - **1.343** (published 07/15/2026, rollout from 07/28/2026): *"When init containers require more resources than the running containers, the difference is reflected as an additional data point, so aggregating at pod scope shows the true effective resource reservation."* Only **pod-scope** aggregations change, and only for pods whose init containers out-request the app containers.
> - **1.345** (published 08/12/2026, staged rollout from 08/25/2026): the same mechanism for **Kubernetes 1.34+ pod-level resources** — and *"For limits, this difference can be negative if the sum of container-level limits exceeds the pod-level limit."* So a pod-scope limit can also go down.
> - **1.347** (published 10/01/2026, staged rollout from 09/30/2026) fixed an *"incorrect calculation of effective pod resource requests and limits when a restartable init container (native sidecar) preceded one or more non-restartable init containers."*
>
> What this means for the thresholds above: the S/M/L tables and routing replica counts are absolute figures and stand as published. The `*_usage` and `*_working_set` metrics measure consumption, not reservation, and are unaffected; so are container-level request/limit series. If you compute request/limit percentages **at pod scope**, re-baseline after the ActiveGate version (and, for 1.345, the cluster's Kubernetes version) changes, rather than comparing a reading across the change.
>
> **Sources:** Sizing guide for ActiveGates in the Kubernetes monitoring use-case (DT docs) — scenario tables, replica counts, OTLP and Log Agent throughput, the replica formula, the split trigger, the two-sets recommendation, the no-CPU-limit preference and the health thresholds are all from this page (re-read 10/02/2026; *"Updated on Jul 15, 2026"*), DynaKube parameters (DT docs) — *"replicas Number of replicas for the Kubernetes monitoring pods. Defaults to one. Set more than one replica for high-availability mode."*, ActiveGate 1.343 release notes (DT docs), ActiveGate 1.345 release notes (DT docs), ActiveGate 1.347 release notes (DT docs) — each quoted above. SaaS 1.343 release notes (DT docs) — the containerized log-ingest sizing quote. **Derived:** "capacity comes from moving up a size" combines the per-size platform table (no replica count) with `replicas` being documented for high availability.

5. Synthetic-Enabled ActiveGates

Synthetic is the one capability where the sizing unit is **work per hour**, and where Dynatrace publishes discrete node sizes. All sizes below sustain up to ~300k HTTP monitor executions per hour — the browser-monitor column is what separates them, because each browser monitor drives a real browser:

Node size	vCPU	RAM	Free disk	Max browser monitor executions/hr	Max high-resource HTTP monitor executions/hr
XS	2	4 GB	20 GB	300	10k
S	4	8 GB	25 GB	650	20k

Node size	vCPU	RAM	Free disk	Max browser monitor executions/hr	Max high-resource HTTP monitor executions/hr
M	8	16 GB	30 GB	1,200	60k
L	16	32 GB	40 GB	2,200	100k

The ~300k HTTP figure applies to ordinary HTTP monitors. A **high-resource** HTTP monitor — one with pre- or post-execution scripts, OAuth2 authorization, or Kerberos authentication — has a far lower ceiling, so an estate heavy in authenticated API checks can be bound by that column rather than by the browser one.

Planning rules that change the architecture, not just the size:

- **Dedicate the ActiveGate.** Synthetic-enabled ActiveGates are *"more demanding"* than routing ActiveGates, support a narrower OS matrix, and *"you can't use one ActiveGate for multiple locations"* — so a synthetic node does synthetic, for exactly one location.
- **At least two nodes per location.** Dynatrace: *"We recommend using at least two ActiveGates for a location. Additional ActiveGates are used for failover and load balancing."* Size the location so the monitor schedule still fits if one node is down.
- **Budget by the browser column.** Convert your monitor plan to executions/hour (monitors × frequency × steps where relevant) and pick the node size with comfortable margin — then scale the *location* by adding nodes rather than pushing one node to its ceiling.

Deployment and location design live in SYNTH-04; containerized synthetic locations on Kubernetes have their own requirements page.

> **Sources:** Requirements for private Synthetic locations (DT docs) — node-size table quoted (*"Estimated maximum number of browser monitor executions/h"*), including the high-resource HTTP monitor row (*"pre- or post-execution scripts, OAuth2 authorization, Kerberos authentication"*); Private Synthetic locations (DT docs) — two-node recommendation and one-location-per-ActiveGate constraint quoted. **Derived:** the executions-per-hour budgeting recipe operationalizes the published per-node ceilings.

6. High Availability, Network Zones, and Placement

Sizing an individual ActiveGate is half the job; the other half is sizing the **group of ActiveGates a zone can fail over across**.

How failover works. Network zones *"prevent inefficient connections by defining the ActiveGate that each OneAgent should use."* When no ActiveGate in the primary zone is reachable, OneAgents try the zone's configured **alternative zones**, then behave per the zone's **fallback mode**: *Any ActiveGate* (default — traffic routes to any available ActiveGate), *Only default zone*, or *None* (traffic stays in the

primary and alternative zones; if no ActiveGate there is available, *"OneAgents don't send data to another zone"*). Two sizing consequences:

- **Survivor capacity is the real requirement** — the requirements page states it directly: remaining ActiveGates must be able to take over the traffic of unavailable ones. If a zone runs two ActiveGates at 45% CPU each and one dies, the survivor takes (roughly) the full load — past the 50% guideline and climbing. In community practice the working rule is **N+1 per zone, where N alone can carry the zone's whole load at ≤50% CPU** — the docs define the failover mechanics but don't publish a per-zone count; verify the headroom against your own self-monitoring numbers (§7).
- **Fallback mode is a capacity decision, not just a security one.** *Any ActiveGate* means a zone failure silently lands its load on ActiveGates you sized for other zones. *None* protects the fleet but means OneAgents send nothing while their zones are down. Pick deliberately, and if you rely on alternative zones, make sure those ActiveGates are both **reachable** from the primary zone's OneAgents (a documented requirement) and **sized** to absorb the addition.

Placement basics (depth in ONBRD-03 and M2S-03): one or more ActiveGates per network zone/datacenter close to the OneAgents they serve; ActiveGate **groups** to control which ActiveGates handle which purpose; and remember that a proxy or firewall between OneAgents and the ActiveGate shapes effective throughput as much as the machine size does.

Size for the egress, not only the ActiveGates. Routing is also what keeps a fleet's connections off your NAT and firewall: Dynatrace says an ActiveGate lets you *"reduce your interaction with Dynatrace to one single point—available locally."* In field experience, a large fleet sent direct instead can hold tens of thousands of connections at one egress device and exhaust its port pool or session table. ONBRD-03 works through the arithmetic. So when a zone's hosts share one internet egress, host count makes the case for routing through ActiveGates, and for sizing them under §3. Being able to reach your environment does not settle it.

> **Sources:** Get started with network zones (DT docs) — failover behavior, alternative zones, the reachability requirement, and fallback modes *"Any ActiveGate (default)"*, *"Only default zone"*, *"None"*; Linux ActiveGate hardware and system requirements (DT docs) — *"The machine running ActiveGate shouldn't exceed 50% CPU and 80% memory."* and *"To ensure high availability, remaining ActiveGates should be able to take over traffic from unavailable ActiveGates."*; Route OneAgent traffic to Dynatrace (DT docs) — *"In this way Dynatrace ActiveGate allows you to reduce your interaction with Dynatrace to one single point—available locally."*

7. Detecting Saturation — Self-Monitoring Signals

Every SaaS and Managed environment ships ActiveGate **self-monitoring metrics** — no setup required. They exist on two surfaces: a Grail-native **Gen3** family, `dt.sfm.active_gate.*`, and a classic **Gen2** family, `dsfm:active_gate.*` (covered at the end of this section). This entry works in the Gen3 namespace throughout — it is what Dynatrace's own diagnostic dashboard queries, it lives in Grail metric buckets (`dt_system_metrics` / `default_metrics`) where DQL `timeseries` reaches it directly, and every key carries fleet-slicing dimensions (`dt.active_gate.id` , `host.name` ,

`dt.active_gate.group.name`, `dt.network_zone.id`). Four signals, watched per ActiveGate, answer "is this ActiveGate the right size" continuously:

Metric	What it tells you	Act when
`dt.sfm.active_gate.jvm.cpu_usage`	ActiveGate Java process CPU	Sustained peaks above the **50%** guideline
`dt.sfm.active_gate.jvm.heap_memory_used` / `…_available`	Heap pressure	Used/available ratio trending past **80%**
`dt.sfm.active_gate.communication.queue.outgoing.usage`	Messages queued faster than they're being sent out	Queue grows and doesn't drain
`dt.sfm.active_gate.communication.messages.dropped`	Messages dropped — data loss	**Any** non-zero value

`dt.sfm.active_gate.communication.agent_modules.connected` rounds out the picture by showing how OneAgent load is actually distributed across the fleet.

```
// ActiveGates whose JVM CPU peaked above the 50% headroom guideline (24h)
timeseries cpu = avg(dt.sfm.active_gate.jvm.cpu_usage), from:-24h, by:{dt.active_gate.id, host.name}
| fieldsAdd peak = arrayMax(cpu)
| filter peak &gt; 50
| sort peak desc
```

```
// Heap utilization per ActiveGate — compare against the 80% memory guideline
timeseries { used = avg(dt.sfm.active_gate.jvm.heap_memory_used), available = avg(dt.sfm.active_gate.jvm.heap_memory_available) }, from:-24h, by:{dt.active_gate.id, host.name}
| fieldsAdd heap_used_pct = round(arrayAvg(used) / arrayAvg(available) * 100, decimals: 1)
| fieldsRemove used, available
| sort heap_used_pct desc
```

```
// Outgoing queue and dropped messages per ActiveGate (7d)
// default: 0 because drop metrics are sparse — they only report when drops occur.
// A healthy AG always has a small queue, so read the columns rather than filtering on
queue &gt; 0:
// any total_dropped is data loss; a queue_trend that stays positive across runs is a
queue that is not draining.
timeseries { queued = avg(dt.sfm.active_gate.communication.queue.outgoing.usage), dropped
= sum(dt.sfm.active_gate.communication.messages.dropped, default: 0) }, from:-7d, by:
{dt.active_gate.id, host.name}, union:true
| fieldsAdd total_dropped = arraySum(dropped), peak_queue = arrayMax(queued), queue_trend
= arrayLast(queued) - arrayFirst(queued)
| fields dt.active_gate.id, host.name, total_dropped, peak_queue, queue_trend
| sort total_dropped desc, peak_queue desc
```

```
// How OneAgent load is distributed across the fleet, per zone (24h)
timeseries agents = avg(dt.sfm.active_gate.communication.agent_modules.connected),
from:-24h, by:{dt.active_gate.id, host.name, dt.network_zone.id}
| fieldsAdd avg_connected = round(arrayAvg(agents), decimals: 0), network_zone =
if(isNull(dt.network_zone.id), "default", else: dt.network_zone.id)
| fieldsRemove dt.network_zone.id
| sort avg_connected desc
```

Wire the first three into anomaly-detection or a scheduled workflow (the ALERT and WFLOW series own the routing patterns) so a saturating ActiveGate pages someone before `messages.dropped` goes non-zero. For containerized ActiveGates, layer on the Kubernetes-native thresholds from §4 (CPU >85% of request, throttling >10%, working set >80%, pipeline >50–60 s).

The full catalog — and the ready-made diagnostic dashboard

The namespace goes far beyond the four core signals, and it is exactly what Dynatrace's own **"ActiveGate diagnostic overview" dashboard** is built on (since ActiveGate 1.345 it lives in Fleet Management — see the note below) — so before hand-building saturation tiles, open or import that dashboard: it already covers host vitals, the Java process, networking, and REST API health, with per-zone/group/instance filtering.

> **Where the dashboard lives changed (ActiveGate 1.345 — staged rollout from 08/25/2026).** Verbatim: *"The **ActiveGate Diagnostic Overview** dashboard has moved from **Dashboards** to **Fleet Management**."* If you go looking for it under Dashboards on a tenant that has taken 1.345 and find nothing, it was moved, not removed — and the same release replaces its ActiveGate ID filtering with Smartscape Node ID (*"It now uses Smartscape Node ID instead of ActiveGate ID for filtering"*). The metrics it charts, and every query in this section, are unaffected.
>
> **A dimension worth having (ActiveGate 1.345).** Verbatim: *"ActiveGate self-monitoring (SFM) metrics now include a host entity dimension when the ActiveGate is monitored by a OneAgent."* That closes a real gap in the queries above: `dt.sfm.active_gate.*` could be sliced by `dt.active_gate.id`, `host.name`, group, and network zone, but not joined to the **host entity** — so ActiveGate saturation and the host vitals underneath it were two separate investigations. Where the ActiveGate host runs a OneAgent, the two now line up in one query. It is additive: nothing above

> breaks on tenants that have not taken 1.345, and the dimension is absent on ActiveGates that no OneAgent monitors.

A live-tenant enumeration (July 2026) shows ~53 keys in the namespace. By area:

Area	Metrics	Notable dimensions / use
Host vitals	`system.cpu_usage`, `system.total_memory` / `system.free_memory`, `storage.volume.free` / `.total`	`storage.volume.mountpoint` / `storage.volume.path` (these replaced the older `volume` dimension in AG 1.335; an ActiveGate still below 1.335 reports `volume`, so group by whichever your fleet emits)
Java process	`jvm.cpu_usage`, `jvm.heap_memory_used` / `_available`, `jvm.gc.duration` (replaces the deprecated `jvm.gc.major_collection_time`), `thread_pool.busy_threads` / `thread_pool.queue_size`	`thread_pool_name`
Networking	`communication.messages.dropped` / `.rejected` / `.resent`, `communication.agent_modules.connected`, `traffic.server.sent` / `.received` (clients ↔ AG), `traffic.client.sent` / `.received` (AG ↔ Dynatrace)	zone / group / instance
Event ingest	`event_ingest.event_incoming_count`, `.drop_count`, `.event_otlp_size` / `.event_json_size`, attribute counters	`drop_reason` (`operation` is documented but null on live data) — incoming_count is the denominator that turns drops into a **drop rate**. Filter on `dt.active_gate.working_mode == "environment"` (see below)
Metrics ingest	`metrics.ingest.otlp.datapoints.received.total` vs `.accepted`, `metrics.ingest.grail.datapoints.accepted`	received minus accepted = OTLP metric datapoints rejected on the way in
REST API	`rest.request_count`, `rest.request_size` / `response_size`, `rest.response_time`	`operation`, `response_code`
Kubernetes module	`kubernetes.pipeline_duration`, `kubernetes.api.query_count` / `.query_duration` / `.connections.pool.available`, `kubernetes.events.observed` / `.processed`, cache size/evictions	`kubernetes.pipeline_duration` is the observable behind §4's "pipeline execution time > 50–60 s" scale-up threshold. **`kubernetes.api.query_count` steps down at the 1.343 upgrade — see the note below.**

Area	Metrics	Notable dimensions / use
AWS module	`aws.data_delay`, `aws.data_lost_monitoring_status`, `aws.elements.bad` / `.reported` / `.total`, `aws.empty_responses`, `aws.query_time`, `aws.requests`	health of the cloud-API-polling capability (§2) — delays/bad elements mean the polling AG is behind, not that AWS is down
Other modules	`rum.beacon_forwarded_count`, `storage.directory.size` / `.limit` (quotas by `storage.volume.mountpoint`), `filecache.size_limit`	

> **Rolling out (ActiveGate 1.343) — `kubernetes.api.query_count` will step *down*, and that is the improvement.** ActiveGate 1.343 (published 07/15/2026, rollout from 07/28/2026) changes the Kubernetes module to call the Kubernetes API **once per minute**, down from multiple calls per collection cycle. So at the upgrade `kubernetes.api.query_count` drops sharply — this is the efficiency change landing, **not** the ActiveGate going quiet or the module failing.
>
> The trap is in the direction of the reading. Falling query counts otherwise look like a module that has stopped collecting, and a rule written to alert on a *drop* will fire on the upgrade. Conversely, a rule reading the post-upgrade level as headroom is comparing against a baseline that no longer exists. **Re-baseline `kubernetes.api.query_count` after the upgrade reaches the cluster before treating its trend as a saturation signal at all** — and correlate against `kubernetes.events.observed` / `.processed` and `kubernetes.pipeline_duration`, which reflect whether work is actually still being done. Verify the ActiveGate version in the cluster to know which side of the change a series is on.

A null zone/group dimension means the default zone/group — coalesce with `if(isNull(…), "default", else: …)` before filtering or charting. Enumerate what your own tenant exposes — the catalog grows with the capabilities you run:

```
// Discover the ActiveGate self-monitoring catalog on your tenant
metrics
| filter startsWith(metric.key, "dt.sfm.active_gate")
| summarize series = count(), by:{metric.key}
| sort metric.key asc
```

One behavior to know: **drop/error metrics are sparse** — a key like `event_ingest.drop_count` only reports (and only enumerates) when drops actually occur. Chart them with `default: 0` (as the diagnostic dashboard does) so "no data" renders as the zero it means, and don't read a key's absence from the discovery query as it not existing.

The dashboard's own coloring thresholds are a useful reference point — slightly more conservative than, and consistent with, the 50%/80% machine guideline:

Signal	Amber at	Red at
System / JVM CPU, memory, heap	≥ 40%	≥ 70%
Storage volumes, directory quotas	≥ 50%	≥ 80%
JVM GC time (share of a 10-min interval)	≥ 5%	≥ 10%

Two signals from that catalog are worth watching beyond the four core signals, because they catch saturation the CPU/heap averages hide:

```
// GC pressure — GC time as a share of each 10-min interval
// The diagnostic dashboard colors this amber at 5%, red at 10%
// jvm.gc.duration replaces the deprecated jvm.gc.major_collection_time; an AG that
// has not taken a newer version may report only the old key — swap it in if an AG is
missing
timeseries gc_time = avg(dt.sfm.active_gate.jvm.gc.duration, rollup:sum), from:-24h,
interval:10m, by:{dt.active_gate.id, host.name}
| fieldsAdd gc_pct = 100 * duration(toLong(gc_time[]), "ms") / interval
| fieldsAdd peak_gc_pct = arrayMax(gc_pct)
| filter peak_gc_pct &gt; 5
| sort peak_gc_pct desc
```

Some pools sit at a constant non-zero queue size on a healthy ActiveGate — on the validation tenant (10/15/2026) Communication Layer held 39 and Register 12 or 25 for the whole day on every ActiveGate — so look for movement above a pool's own floor, not for a value above zero:

```
// Thread-pool backpressure — a queue that rises above its own floor is a pool that can't
keep up.
// Some pools sit at a constant non-zero value on a healthy ActiveGate, so test movement,
not &gt; 0.
timeseries queue_size = max(dt.sfm.active_gate.thread_pool.queue_size), from:-24h, by:
{dt.active_gate.id, host.name, thread_pool_name}
| fieldsAdd floor = arrayMin(queue_size), peak_queue = arrayMax(queue_size), queue_trend =
arrayLast(queue_size) - arrayFirst(queue_size)
| filter peak_queue &gt; floor
| fields dt.active_gate.id, host.name, thread_pool_name, floor, peak_queue, queue_trend
| sort queue_trend desc, peak_queue desc
```

And for the §6 HA math, a one-query fleet inventory — how many ActiveGates each zone actually has:

```
// ActiveGates per network zone — the denominator for the survivor-capacity check
timeseries cpu_samples = count(dt.sfm.active_gate.system.cpu_usage), from:-24h, by:
{dt.network_zone.id, dt.active_gate.id}
| fieldsAdd network_zone = if(isNull(dt.network_zone.id), "default", else:
dt.network_zone.id)
| summarize total_active_gates = countDistinctExact(dt.active_gate.id), by:{network_zone}
```

Pinpointing ingest loss by endpoint and reason

The namespace's sharpest diagnostic — and one the ready-made dashboard does *not* chart — turns "the ActiveGate dropped something" into "*this* endpoint dropped *this many* events for *this* reason." The load-bearing metric is `dt.sfm.active_gate.event_ingest.drop_count`, dimensioned by:

Dimension	What it splits by	Why it matters
`operation`	Documented by the diagnostic dashboard as the ingest endpoint (e.g. `POST /otlp/v1/logs`), but **null on live data** (verified 08/27 and 09/28/2026) — don't filter on it	Would isolate OTel log loss from other ingest paths, if populated
`drop_reason`	`head_on_queue_limit` (in-memory ingest queue full — events dropped at the head) vs `disk_queue_limit` (disk buffer exhausted); `unexpected_exception` also occurs (259 drops over 7 days on the validation tenant, 10/02/2026)	Tells you *which* buffer saturated, which changes the fix
`dt.active_gate.id`, `host.name`, `dt.active_gate.group.name`, `dt.network_zone.id`, `dt.active_gate.working_mode`	Fleet slicing	Pinpoints the ActiveGate and zone. `working_mode` also separates your own ActiveGates (`environment`) from `embedded` series

```
// Ingest drops at the ActiveGate, by machine and drop reason (24h)
//
// Do NOT scope this with `| filter operation == "POST /otlp/v1/logs"`. The
// `operation` dimension is null on every data point of this metric (verified
// 08/27/2026: 73 real drops in 24h, all with operation null), so that filter
// discards the only rows the query exists to surface and reports "no loss"
// while records are being dropped. `drop_reason` carries the diagnostic signal.
// working_mode == "environment" keeps your own ActiveGates and drops the `embedded`
series.
timeseries drop_count = sum(dt.sfm.active_gate.event_ingest.drop_count), from:-24h, by:
{dt.active_gate.id, host.name, drop_reason}, filter:{dt.system.bucket ==
"dt_system_metrics" AND dt.active_gate.working_mode == "environment"}
| fieldsAdd total_dropped = arraySum(drop_count)
| filter total_dropped &gt; 0
| sort total_dropped desc
```

```
// Average OTLP event payload size per ActiveGate and signal type (24h)
// Rising payload size consumes routing capacity faster than the host-count tables assume
(§3)
// Returns no rows if none of your ActiveGates received OTLP events in the window
timeseries otlp_bytes = avg(dt.sfm.active_gate.event_ingest.event_otlp_size), from:-24h,
by:{dt.active_gate.id, host.name, event.type}, filter:{dt.system.bucket ==
"dt_system_metrics" AND dt.active_gate.working_mode == "environment"}
| fieldsAdd avg_event_bytes = round(arrayAvg(otlp_bytes), decimals: 0)
| sort avg_event_bytes desc
```

Reading the `drop_reason` split: `head_on_queue_limit` is a throughput problem — the ActiveGate is receiving faster than it can process, so scale up/out or split capabilities (§8). `disk_queue_limit` means the disk buffer that absorbs uplink outages filled up — check connectivity to Dynatrace first, then the buffer allowance (the OPLOGS series documents the ActiveGate disk-queue behavior and its default size cap for log delivery). A companion metric, `dt.sfm.active_gate.communication.queue.outgoing.usage`, gives the outgoing-queue signal already used in the core queries above.

To judge severity rather than raw counts, pair `drop_count` against `event_ingest.event_incoming_count` on the same dimensions — a thousand drops out of a thousand incoming events is an outage, out of a billion it's noise.

Filter the `event_ingest.*` family to your own ActiveGates. On SaaS these series are also reported with `dt.active_gate.working_mode == "embedded"` from hosts that are not ActiveGates you run (no `ACTIVEGATE` Smartscape node, host names under `internal.dynatrace.com`). On the validation tenant on 10/02/2026 they carried 28.4 million of 28.5 million incoming events over 7 days, and an unfiltered payload-size query returned only those hosts. The docs do not describe the `embedded` mode; in community practice it is read as Dynatrace-side ingest, so keep `dt.active_gate.working_mode == "environment"` in any fleet sizing or drop-rate query, as the two above do.

The classic (Gen2) counterpart — `dsfm:`

The `dsfm:active_gate.*` family is the **Metrics Classic** form of the same self-monitoring signals — its reference page sits in the Metrics Classic docs section, and it's built for the Gen2 surfaces: metric selectors, Data Explorer, and the classic Metrics API. Two reasons it still matters even though this section queries Gen3 throughout:

- **It's the documented one.** The ActiveGate self-monitoring docs page lists `dsfm:` keys only. Of the `dt.sfm.*` keys used above, two appear in the Kubernetes sizing guide's validation table (`kubernetes.pipeline_duration` and `jvm.gc.major_collection_time`); the rest are undocumented. Their provenance is the shipped diagnostic dashboard plus live-tenant verification (including a full key enumeration), so confirm keys and dimension values in your own tenant before wiring alerts to them. The sfm page also marks *"dsfm:active_gate .jvm .gc .major_collection_time … Deprecated. To be removed. Use: dsfm:active_gate.jvm.gc.duration"* — which is why the GC query above reads `jvm.gc.duration`. The docs' authoritative signal interpretations — for the queue, *"growing value indicates messages received faster than sent out"*; for drops, *"non-zero value may indicate data loss"* — apply to both forms.

- **Classic surfaces still consume it.** Anything driven by a metric selector rather than DQL reads the `dsfm:` form — and on **Dynatrace Managed**, where Grail isn't available, `dsfm:` is the self-monitoring surface, full stop.

If you do query `dsfm:` keys in DQL, mind the colon: `timeseries` parses an unquoted `dsfm:` prefix as a parameter name, so backtick-quote the key — `avg(`dsfm:active_gate.jvm.cpu_usage`)`.

Provenance for this section: the Gen2/Gen3 labeling reads the docs' Metrics-Classic placement of `dsfm:` against `dt.sfm.*`'s Grail-bucket residency — Dynatrace does not state the generation split explicitly. The `dt.sfm.active_gate.*` catalog comes from a live-tenant key enumeration (July 2026, ~53 keys) plus the shipped diagnostic dashboard's own definitions (threshold colorings, `drop_reason` / `operation` dimension values), none of it yet on the sfm documentation page. Every query here was syntax-validated against a live tenant; the GC, queue/drop and event-ingest queries were re-executed 10/02/2026 with no notifications.

> **Sources:**
>
> - ActiveGate self-monitoring metrics (DT docs) — `dsfm:` metric keys, descriptions, the queued/dropped interpretations, and the `major_collection_time` deprecation quoted;
> - Self-monitoring metrics (DT docs) — the `dsfm:` family's placement under Metrics Classic and its selector/Data-Explorer query surface;
> - Sizing guide for ActiveGates in the Kubernetes monitoring use-case (DT docs) — lists *"dt.sfm.active_gate.kubernetes.pipeline_duration"* and *"dt.sfm.active_gate.jvm.gc.major_collection_time"* among its validation metrics;
> - ActiveGate 1.345 release notes (DT docs) — the Diagnostic Overview dashboard's move to Fleet Management, and SFM metrics gaining a host entity dimension when the ActiveGate is OneAgent-monitored;
> - Linux ActiveGate hardware and system requirements (DT docs) — 50%/80% thresholds
> - ActiveGate 1.335 release notes (DT docs) — *"storage.volume.path and storage.volume.mountpoint now replace the previous volume dimension."* (re-read 10/22/2026; the 1.347 notes, updated 10/16/2026, no longer list a disk-space metric removal)
> - ActiveGate 1.343 release notes (DT docs) — the Kubernetes module now calls the Kubernetes API once per minute, down from multiple calls per collection cycle
> - **Derived:** the Gen2/Gen3 split, the 50%/80% thresholds applied to JVM-level metrics, and the `dt.sfm.active_gate.*` catalog are all inferred rather than documented on the sfm page

8. Scale Up or Scale Out?

When §7 says an ActiveGate is running hot, the right move depends on which capability is saturating:

Signal	Capability	Move
CPU peaks past 50% on a routing ActiveGate	Routing	**Scale out** — add an ActiveGate to the zone; the addition also buys the failover headroom the requirements page asks for (§6)
Kubernetes-monitoring pod hits the §4 thresholds (working set >80%, pipeline >50–60 s)	K8s platform monitoring	**Scale up** — move to the next scenario's requests/limits; the guide gives no replica count for this workload, and `replicas` is documented for high availability (§4)
In-cluster routing replicas saturating	K8s OneAgent routing	**Scale out** — raise replicas per the §4 routing table (for OTLP log ingest, the §4 replica formula); split into separate deployments past *"CPU usage >80%, memory usage >85%, or frequent GC pauses"*
Browser-monitor queue delays / executions near the node ceiling	Synthetic	**Scale up the node** (XS→S→M→L) or **add nodes to the location** — the location load-balances across nodes
Outgoing queue grows while the network is healthy	Any	**Scale up** CPU and memory first; if one ActiveGate carries several jobs, **split the capabilities** onto dedicated ActiveGates (§2)
One ActiveGate accumulating jobs over time	Mixed	**Split before you supersize** — a purpose-per-ActiveGate fleet is easier to size, update (FAQ-05), and troubleshoot

The last two table rows (scale up first; split before you supersize) are community practice rather than documented rules. In community practice, the general shape is: **routing scales out, monitoring workloads scale up, synthetic scales by nodes** — and capability separation is the escape hatch when one box is doing too many different things for either axis to fix.

> **Sources:** Sizing guide for ActiveGates in the Kubernetes monitoring use-case (DT docs) — platform-monitoring vs routing split, *"Scale horizontally : Increase the number of ActiveGate replicas"* / *"Scale vertically : Increase CPU and memory resource requests"*; Private Synthetic locations (DT docs) — node failover/load-balancing; Get started with network zones (DT docs) — zone failover.

9. Recommended Approach

1. **Inventory the jobs first (§2).** List every capability each planned or existing ActiveGate carries: routing, Kubernetes monitoring, extensions, cloud API polling, synthetic, z/OS. If one ActiveGate carries more than routing plus light extensions, plan a split.
2. **Quantify each job's driver.** Hosts per zone for routing; pods per cluster for Kubernetes; browser executions/hour for synthetic; extension instances for EEC.
3. **Pick baselines from the current tables (§3–§5)** — and re-check the docs pages when you do, because the reference instances and estimates move between versions.

4. **Apply the HA multiplier (§6).** Per zone: enough ActiveGates that the fleet holds ≤50% CPU with one member down. Per synthetic location: at least two nodes.
5. **Deploy per the owning series** — ONBRD-03 for host-based install/groups/zones, K8S for DynaKube, SYNTH-04 for locations.
6. **Baseline the self-monitoring metrics on day one (§7)** and alert on CPU >50%, heap >80%, queue growth, and any dropped messages.
7. **Scale on signal, not on schedule (§8):** out for routing, up for Kubernetes monitoring, nodes for synthetic — and split capabilities when one ActiveGate does too much.
8. **Re-review at growth events:** onboarding waves, new clusters, monitor-schedule expansions, and before each migration phase (M2S-03/S2S-04 both put ActiveGate re-sizing in the design step for a reason).

10. Common Gotchas

Symptom	Likely cause	Where
Data gaps during an ActiveGate outage	Zone sized for average load, not survivor load — the remaining ActiveGates saturated	§6
`messages.dropped` non-zero but CPU looks fine	Outgoing queue backed up — uplink/proxy bottleneck, not machine size	§7
OTel logs going missing but the fleet-level dropped-message metrics look clean	Endpoint-level ingest drops — check `dt.sfm.active_gate.event_ingest.drop_count` by `drop_reason` per ActiveGate (`operation` is currently null) (`head_on_queue_limit` = throughput, `disk_queue_limit` = uplink/buffer)	§7
Processing lags but JVM CPU looks moderate	GC pressure — GC time above ~10% of the interval means the heap is too small for the workload	§7
Browser monitors intermittently delayed	Synthetic node at its executions/hour ceiling, or location has a single node	§5
Kubernetes-monitoring data lags on a big cluster	ActiveGate CPU-throttled by its limit, or pipeline time past 50–60 s	§4
One ActiveGate monitoring several clusters misbehaves	The sizing guide's tables are per cluster; in community practice one Kubernetes-monitoring ActiveGate per cluster is the easier pattern to size	§4
EEC logs lost during an outage	EEC log persistence turned itself off — less than 2,136 MB free disk, so logs are sent without the reliability buffer	§3
"The table said 800 hosts but we saturated at 500"	Estimates assume typical per-host volume — heavy log routing or dense instrumentation consumes it faster	§3, §7

Symptom	Likely cause	Where
Everything degraded after adding one more job to a shared ActiveGate	Capability stacking — demanding jobs deserve dedicated ActiveGates	§2, §8
Zone failure overloaded a *different* zone's ActiveGates	Fallback mode *Any ActiveGate* rerouted the load somewhere unsized for it	§6
Sites lose internet access after a fleet-wide OneAgent rollout without ActiveGates	Every host's OneAgent connections pass through one NAT/firewall egress and exhaust its port pool or session table	§6, ONBRD-03

> **Sources:** each row maps to the documented constraint or threshold cited in the referenced section — see the Sources blocks of §3–§8.

> ⚠️ **Disclaimer:** This content is AI-generated, community-driven, and **not supported by Dynatrace**. ActiveGate sizing tables, supported platforms, and self-monitoring metrics evolve across releases — always verify current numbers against the Dynatrace documentation and your own tenant's self-monitoring metrics before committing hardware.